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Case study

Concrete compressive strength prediction using an explainable boosting machine model

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ABSTRACT

The mixing ratio of the raw materials has a significant impact on concrete compressive strength. Although the compressive strength of concrete can be inferred from the mix ratio, it is frequently challenging to determine how each mix ratio parameter affects the results. In this study, an Explainable Boosting Machine (EBM) is applied to predict the concrete compressive strength and explain the contribution of mix ratio factors on the compressive strength. Meanwhile, the combined algorithm selection and hyperparameter optimization problem in machine learning is implemented by employing a Bayesian optimization technique. A dataset consisting of 1030 concrete compressive strength test data has been used for model building. The results show that the Bayesian optimization algorithm iteratively constructs the algorithmic/hyperparametric model and identifies the optimal point in the space, significantly reducing the time consumption in the ML model building process. In terms of model prediction performance, the EBM algorithm shows excellent results with $R^2 = 0.93$, RMSE = 4.33, and MAE = 3.10. The EBM algorithm allows one to fully interpret the contribution of individual features to the prediction results from both global and local aspects, by which the impact of each mix ratio parameter on the compression strength of the concrete can be further determined.

1. Introduction

Concrete is the most widely used building material and has an essential impact in the civil engineering industry. It is mainly used in building structures to support loads and resist pressure, whereas compressive strength is the most significant indicator of concrete quality. The 28-day compressive strength of concrete is a key parameter when designing reinforced concrete structures. Current test methods for measuring the compressive properties of concrete are based on artificial measurements. The concrete specimens used are usually the same proportion concrete specimens reserved at the time of casting on site, or by coring specimens from the structure after hardening. Despite the high accuracy of this method, the process is complicated and may cause irreversible damage to the building during the measurement process.

The compressive strength of concrete is extremely dependent on its mixing compositions [1]. The amount of cement and supplementary cementitious materials (SCMs), the ratio of water to cement (W/C), the particle size of fine and coarse aggregates, and the type and quantity of chemical admixtures are factors that affect the concrete compressive strength. Meanwhile, the effects of various raw materials are not completely independent, there is an interaction between them. Different exposed conditions, such as the temperature and relative humidity of the air, also affect the strength characteristics of the concrete [2,3]. Therefore, the composition of

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mixtures and different exposed conditions lead to a complex interrelationship, which causes a non-linear development of the compressive strength and makes the prediction of compressive strength even more difficult [4].

Owing to the complex factors that govern the development of compressive strength, the use of machine learning (ML) models to predict compressive strength is becoming increasingly popular. ML algorithms are capable of predicting compressive strength (output variable) from the type and quantity of mixing components (input variables). ML models can be trained from the gathered dataset that includes input and output variables and figure out the connection between the output and input variables without being constrained by past intuitive understanding [4,5]. In the field of civil engineering, researchers have tried to apply ML methods to predict various properties of structural concrete, such as compressive strength, shear strength, and flexural strength, among others [6–14]. Additionally, a variety of machine learning techniques, including Multiple Linear Regression (MLR), artificial neural network (ANN), support vector machine (SVM), decision Trees (DTs), random forest (RF), adaptive boosting (AdaBoost), and extreme gradient boosting (XGBoost), have been increasingly used [8,15–23]. Although the application of the above ML algorithms has demonstrated its unique advantages in predicting the basic performance of concrete structures, their performance and application have been questioned due to the lack of explainability [24]. The machine learning model seems to the common user to be a 'black box' that output a decision result with a given input. Therefore, it is difficult to know exactly the basis of the decision and whether the decision is reliable. The lack of explainability may lead to serious threats in practical tasks, especially in sensitive tasks related to structural safety. To improve the explainability of models, some scholars have used feature importance analysis to explain the prediction of the model [25]. However, the method of feature importance analysis only provides the importance ranking of features but cannot explain how each feature affects the prediction specifically.

Common improvement in predictive ability of ML models is accompanied by increased model complexity and reduced interpretability. Complex state-of-the-art (SOTA) models, such as XGBoost and RF, are only capable to give feature importance in the global prospective (global explanation). However, it is so difficult to determine how a feature relates to the final output. Such as, whether these factors are positively correlated, negatively correlated, or otherwise more complex relationship, since it only computes the general influence of the features on the output within the overall database context. In addition, feature importance cannot distinguish between the quantitative effect of each feature on each individual sample's predictive value (local explanation). More recently, some explainable machine learning models have been focused on explaining the dependence of prediction results on input variables, such as SHapley Additive exPlanations (SHAP) [26], Local Interpretable Model-agnostic Explanations (LIME) [27]. However, such methods usually require additional interpretation procedures other than prediction models. The SHAP model often needs to compute a large number of Shapley values in the samples, and the LIME model requires predictions for the perturbed samples, either of which greatly reduces the efficiency of the models. Additionally, these techniques are intended to explain model predictions rather than model decision-making process, which may result in inaccurate explanations [24].

A novel interpretable ML method, Explainable Boosting Machine (EBM) [28] has recently been proposed in an attempt to explain the relationship between input and output variables during the prediction process. Previous studies [29–33] demonstrated that the accuracy of EBM is comparable to the state-of-the-art ML approaches, like XGBoost and RF, as it adopts an advanced ensemble learning strategy to train the model. EBM remains highly intelligible and explainable based on generalized additive models [34]. It could provide intrinsic interpretability with the assistance of shape functions constructed for individual features or pairwise interactions. Thus, the decision-making procedure of EBM and final predictions can be interpreted in both global and local aspects without bells and whistles or any additional procedures, which indicates its superiority over complex black box models. EBM has been successfully applied to environmental assessment [29], geoengineering [30],and health care [32], but it has not been used in civil engineering, much less for concrete compressive strength prediction.

The model selection and hyperparameter optimization problem are obstacles for using ML to analyze the compressive strength of concrete. The problem of model selection and hyperparameter optimization has been extensively reported separately [35–37]. However, making the appropriate decision is still difficult when confronted with these freedom degrees. Most selections are driven by the reputation or intuitive appeal of the algorithm and tuning the hyperparameters through time-consuming human intervention. To address these issues, a practical and fully automated approach is proposed to solve the combined algorithm selection and hyperparameter optimization (CASH) problem [38]. CASH problem is solved by using the Bayesian optimization (BO) technique, which iterates to construct models of the algorithmic/hyperparametric landscape and uses these models to find the optimal points. In the CASH problem, the selection of algorithms is also considered as a hyperparameter, thereby achieving the purpose of selecting the optimal combined algorithm. With respect to building a high-quality and practical model for predicting the concrete compressive strength under multi-factor influence, the problem is formulated as a CASH problem and solved by Bayesian optimization in this study.

The overall goal of this study is to propose an explainable ML model that can forecast the compressive strength of concrete. The model combines the powerful nonlinear fitting ability of ML with high interpretability. Bayesian optimization is utilized to solve the CASH problem automatically without manual intervention. To confirm the efficiency of the suggested model, the prediction results are compared with other ML models. The proposed algorithm framework is helpful in achieving the prediction of concrete compressive strength under different mix ratio conditions in a simple and quick manner.

2. Significance of the subject

The mix design is a difficult task due to the non-linear interaction between concrete components and its mechanical properties. In the case of concrete compounded with a variety of supplementary cementitious materials and chemical admixtures, a large number of experimental trials are required to ensure that the performance meets the design requirements. In this study, ordinary Portland cement (OPC), fly ash (FA), slag, water, superplasticizer (SP), coarse aggregate (CA), fine aggregate (FA) and age of concrete cure (Age) are

adopted as input features to predict the concrete compressive strength. Machine Learning methods have great potential for this multidimensional problem, which is able to depict the non-linear relationship between eight parameters and the concrete compressive strength.

3. Predictive model for concrete compressive strength as a CSAH problem

In this study, the proposed prediction model for concrete compressive strength involves simultaneously selecting an ML model and setting its hyperparameters. It is treated as a CASH problem and is solved through a completely automated approach based on Bayesian optimization. For comparison, three typical ML models for tabular data prediction problems are chosen, including a single model Decision Tree (DT), an ensemble model Random Forest (RF), and a gradient boosting tree model XGBoost. Although there are many ML methods, it should be emphasized that the RF and XGBoost ensemble models have consistently demonstrated state-of-the-art performance in a wide range of tasks [39,40]. Therefore, three representative methods are selected in this study for comparative analysis. The details of the proposed model are illustrated in this section.

3.1. Formulation of the CASH problem

Given a dataset $\mathscr{D} \in \mathscr{X} \times \mathscr{Y}$, the prediction of concrete compressive strength is to find a function (initialized by an algorithm A) $f : \mathscr{X} \mapsto \mathscr{Y}$. The learning algorithm A is configured by hyperparameters $\boldsymbol{\lambda} \in \boldsymbol{\Lambda}$, which changes the working mechanism of the learning algorithm A_λ itself [38]. The purpose of the modeling selection of training data $\mathscr{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$ is to decide $A^* \in \mathscr{A}$ with the optimal generalizability, which can be written as:

$$A^* \in \underset{A \in \mathscr{A}}{\operatorname{argmax}} \frac{1}{k} \sum_{i=1}^{k} \mathscr{J}\left(A, \mathscr{D}_{\text{train}}^{(i)}, \mathscr{D}_{\text{valid}}^{(i)}\right) \tag{1}$$

where $\mathscr{D}_{\text{train}}^{(i)}$ and $\mathscr{D}_{\text{valid}}^{(i)}$ are disjoint training and validation sets; k denotes the folds in cross validation; $\mathscr{J}\left(A, \mathscr{D}_{\text{train}}^{(i)}, \mathscr{D}_{\text{valid}}^{(i)}\right)$ is the objective value evaluated on $\mathscr{D}_{\text{valid}}^{(i)}$ to be maximized (or loss to be minimized). In the structured space $\boldsymbol{\Lambda}$, the problem of optimizing hyperparameters can be described as follows.

$$\boldsymbol{\lambda}^* \in \underset{\boldsymbol{\lambda} \in \boldsymbol{\Lambda}}{\operatorname{argmin}} \frac{1}{k} \sum_{i=1}^{k} \mathscr{J}\left(A_\lambda, \mathscr{D}_{\text{train}}^{(i)}, \mathscr{D}_{\text{valid}}^{(i)}\right) \tag{2}$$

The CASH problem can be described as

$$A_{\lambda^*}^* \in \underset{A^{(j)} \in \mathscr{A}, \lambda \in \boldsymbol{\Lambda}^{(j)}}{\operatorname{argmin}} \frac{1}{k} \sum_{i=1}^{k} \mathscr{J}\left(A_\lambda^{(j)}, \mathscr{D}_{\text{train}}^{(i)}, \mathscr{D}_{\text{valid}}^{(i)}\right) \tag{3}$$

In other words, the prediction of concrete compressive strength is solved by Eq. (3) instead of Eqs. (1) and (2) separately, which sets a single optimization problem of combined hierarchical hyperparameters with respect to parameter space $\boldsymbol{\Lambda} = \boldsymbol{\Lambda}^{(1)} \cup \cdots \cup \boldsymbol{\Lambda}^{(k)} \cup \{\lambda_r\}$. This hierarchical hyperparameter optimization is automatedly accomplished by Bayesian optimization and is much efficient and accurate compared with separate procedures.

3.2. Bayesian optimization

Bayesian optimization involves two critical components: a probabilistic proxy model and an acquisition function that determines which point to be further evaluated. The surrogate model is adjusted to all currently observable targets with each iteration. The usefulness of various candidate points is then determined by the acquisition function using the probabilistic model's prediction distribution. The acquisition function is computationally inexpensive compared to evaluating expensive 'black box' functions and therefore can be thoroughly optimized.

In this study, positive expected improvement (EI), denoted as $I(\lambda) = \max\{c - \mathscr{J}_{\max}, 0\}$, is utilized as the acquisition function. The expectation is as follows.

$$\mathbb{E}[I(\boldsymbol{\lambda})] := \int_{\mathscr{J}_{\max}}^{+\infty} \max\{c - \mathscr{J}_{\max}, 0\} \cdot p(\mathscr{J} \mid \lambda) d\mathscr{J} \tag{4}$$

where $p(\mathscr{J} \mid \lambda)$ is obtained by surrogate model Tree Parzen Estimator (TPE). TPE adopts individual models for $p(\mathscr{J})$ and $p(\lambda \mid \mathscr{J})$ to get $p(\mathscr{J} \mid \lambda)$. The calculation is expressed as

$$p(\boldsymbol{\lambda} \mid \mathscr{J}) = \left\{ \begin{array}{l} ampampl(\boldsymbol{\lambda}), \text{if} \mathscr{J} < \mathscr{J}^* \\ \& g(\boldsymbol{\lambda}), \text{if} \mathscr{J} \geq \mathscr{J}^* \end{array} \right\} \tag{5}$$

$$p(\mathscr{J} \mid \lambda) = \frac{p(\boldsymbol{\lambda} \mid \mathscr{J}) p(\mathscr{J})}{p(\boldsymbol{\lambda})} = \frac{l(\boldsymbol{\lambda}) p(\mathscr{J})}{\gamma l(\boldsymbol{\lambda}) + (1 - \gamma) g(\boldsymbol{\lambda})} \tag{6}$$

where, $\mathscr{J}^*$ is chosen as the γ-quantile of the objective values (0.15 by default); $\ell(\lambda)$ and $g(\lambda)$ are density estimates with corresponding objective values smaller or greater than $\mathscr{J}^*$ respectively. The bestλ is the hyperparameter set to maximize Eq. (4).

Finally, hyperparameters of the predictive model for concrete compressive strength as stated in Eq. (3) are optimized by Bayesian optimization, which randomly generates innumerable candidate hyperparameter configurations and selects λ as the minimum value of Eq. (4).

4. ML models for concrete compressive strength prediction

4.1. Explainable boosting machine

EBM s an inherent interpretable model base upon Generalized Additive Model (GAM) with additional pairwise interactions [28]. For specimen (x_i, y_i) in the concrete compressive strength dataset, the prediction $\widehat{y}_i$ is expressed as

$$\widehat{y}_i = \beta_0 + \sum_{j=1}^{p} f_j(x_{ij}) + \sum_{k=1}^{K} f_k(x_{ik_1}, x_{ik_2}) \tag{7}$$

where, β_0 is the intercept, f_js and f_ks are the shape functions. K pairs of features (x_{ik_1}, x_{ik_2}) are chosen greedily by FAST [41]. The overall mean value of f_j is set to 0 since any offset can be added to f_j while minus it from β_0 or other shape functions. As each form functionf_j only affects one feature x_j, the f_j can be completely plotted. The generation, learning and gradient update of shallow decision trees are carried out using only one predictor variable with a low learning rate at the same time. This allows EBM to be interpreted, as prediction is a summary of the contribution of different features and can be visualized via 2D display. EBM uses ensemble techniques to represent f_js as a sequence of trees and improves prediction accuracy by tree-based integration algorithm.

4.2. Decision trees

Decision trees (DTs) is a non-parametric guided method for solving classification and regression problems. It divides the feature space into a series of simple parts, which in turn creates a learning model and uses the simple decision rules inferred from the training data to make predictions. DTs is one of the most popular machine learning algorithms given their intelligibility and simplicity. In this study, DTs serves as the baseline for the evaluation of model performance compared to other ensemble models.

4.3. Random forests

Random Forests (RF) is an integrated model consisting of a combination of the bagging method and decision trees. It combines M randomized DTs and aggregates their predictions by averaging. The j-th tree estimate takes the form

$$T_n(\mathbf{x}; \Theta_j, \mathscr{D}_n) = \sum_{i \in \mathscr{D}_n^\star(\Theta_j)} \frac{\mathbb{1}_{\mathbf{x}_i \in A_n(\mathbf{x}; \Theta_j, \mathscr{D}_n)} Y_i}{N_n(\mathbf{x}; \Theta_j, \mathscr{D}_n)} \tag{8}$$

where Θ_j is the model parameter, $\mathscr{D}_n^\star(\Theta_j)$ is the set of data points selected prior to the tree construction, $A_n(\mathbf{x}; \Theta_j, \mathscr{D}_n)$ is the cell containing $\mathbf{x}$, and $N_n(\mathbf{x}; \Theta_j, \mathscr{D}_n)$ is the number of (preselected) points that fall into $A_n(\mathbf{x}; \Theta_j, \mathscr{D}_n)$.

With M DT models established, the RF prediction $\widehat{y}_i$ of specimen $(\mathbf{x}_i, y_i)$ is

$$\widehat{y}_i = \frac{1}{M} \sum_{M=j}^{M} T_n(\mathbf{x}; \Theta_j, \mathscr{D}_n) \tag{9}$$

4.4. XGBoost

XGBoost is also an integrated learning method, which is an extended form of gradient boosted decision tree (GBDT). Compared to other models, XGBoost is optimized in terms of loss function and loss optimization process [42]. XGBoost runs with a parallel tree boost and can quickly and accurately resolve many data science tasks, achieving state-of-the-art results in many machine learning challenges [43]. XGBoost minimizes the following rule loss function to learn the weak learner set in the integrated model.

$$\mathscr{L} = \sum_i l(\widehat{y}_i, y_i) + \sum_k \left(\gamma N + \frac{1}{2} \nu \|\beta_k\|^2 \right) \tag{10}$$

where, l is the difference between the prediction $\widehat{y}_i$ and the target y_i; N is tree node number; γ and ν are penalty coefficients; $\beta_k$is leaf scores. The first term typically adopts a squared loss function, while the second term is based on the leaf's L_2 norm and the number of tree nodes N. The prediction of XGBoost for concrete is obtained by unweighted summation of all weak learners. All the ML models adopted in this study are illustrated in Fig. 1.

5. Database

5.1. Experimental procedure

A dataset of 1030 concrete compressive strength test results have been utilized for model creation in order to forecast the performance of ML models. The dataset was derived from concrete mix design data from the UCI (University of California Institute) ML Repository, collected by Yeh [44,45], which has been utilized to train and test individual machine learning algorithms [46–49]. The composition of the data consists of eight independent variables and one dependent variable. Independent variables include ordinary Portland cement (OPC), fly ash (FA), slag, water, superplasticizer (SP), coarse aggregate (CA), fine aggregate (FA) and age of concrete cure (Age). The dependent variable is the compressive strength of the concrete. Detailed statistics of each independent and dependent variable of the data set, such as name and unit, minimum/maximum value, mean and standard derivation (SD) of the experimental parameters, are summarized in Table 1. The histograms of all the input parameters and compressive strength are shown in Fig. 2. The histograms show that Portland cement, water, coarse aggregate, fine aggregate, and compressive strength mainly follow Gaussian distributions. With the exception of samples lacking this feature, slag, fly ash, and superplasticizer generally follow Gaussian distributions. Most samples are in the range of 0–28 from the curing age histogram. A broader feature distribution range is used to enhance the machine learning model's generalization ability. The ability of the machine learning model to predict the compressive strength of various concrete components is potentially expanded by large feature distributions.

5.2. Performance measures

The performance of the prediction algorithms used in this study was examined by taking regular performance measurements using three metrics: correlation of determination (R^2), mean absolute error (*MAE*), and root mean squared error (*RMSE*). R^2 denotes the extent of the linear correlation between the actual and predicted results. The value of R^2 ranges from zero to one. As R^2 is closer to one, the better the performance exhibited by the prediction model. The *RMSE*, which symbolizes the difference between the predicted and actual values, is defined as the square root of the average mean square errors. The *MAE* is a statistical measure of prediction accuracy that reveals the error between predicted and actual values. The prediction method performs better when the *RMSE* and *MAE* values are lower. The three measures are listed as follows:

Correlation of determination (R^2)

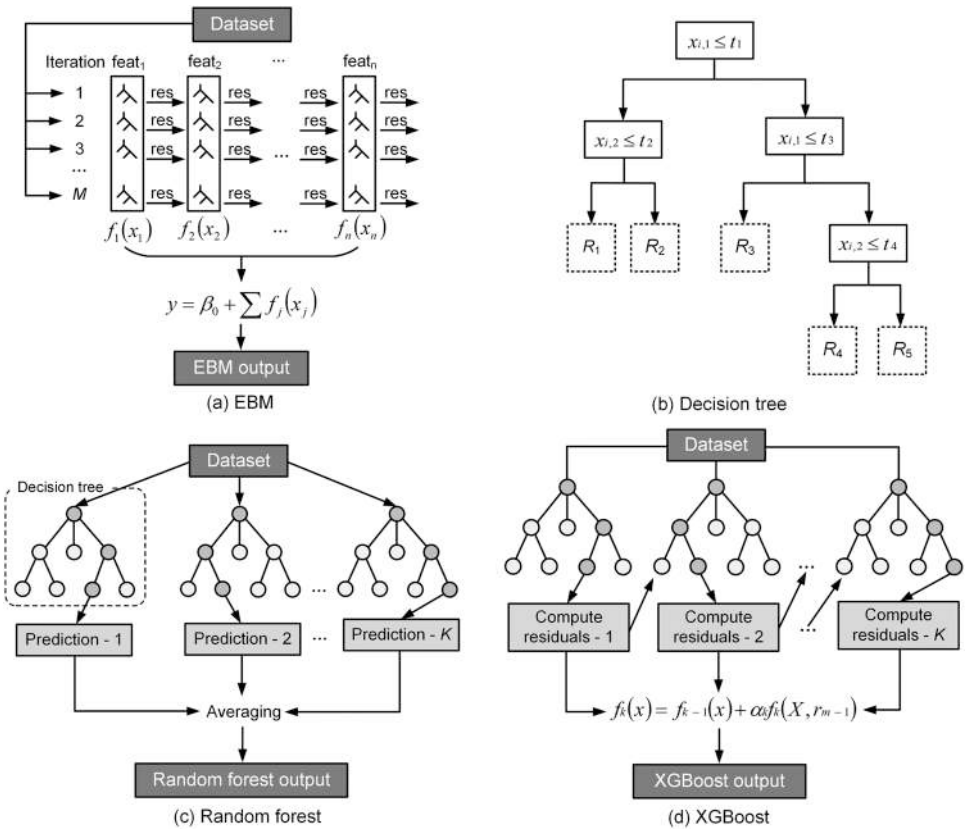

Fig. 1. Schematic illustration of the utilized ML models.

Table 1
Detailed statistics of variables in the dataset.

	Unit	Mean	St.D.	Min.	Med.	Max.
OPC	kg/m^3	281.17	104.51	102.00	272.90	540.00
FA	kg/m^3	54.19	64.00	0.00	0.00	200.10
BFS	kg/m^3	73.90	86.28	0.00	22.00	359.40
Water	kg/m^3	181.57	21.36	121.75	185.00	247.00
SP	kg/m^3	6.20	5.97	0.00	6.35	32.20
CA	kg/m^3	972.92	77.75	801.00	968.00	1145.00
FA	kg/m^3	773.58	80.18	594.00	779.51	992.60
Age	days	45.66	63.17	1.00	28.00	365.00
Strength	MPa	35.82	16.71	2.33	34.44	82.60

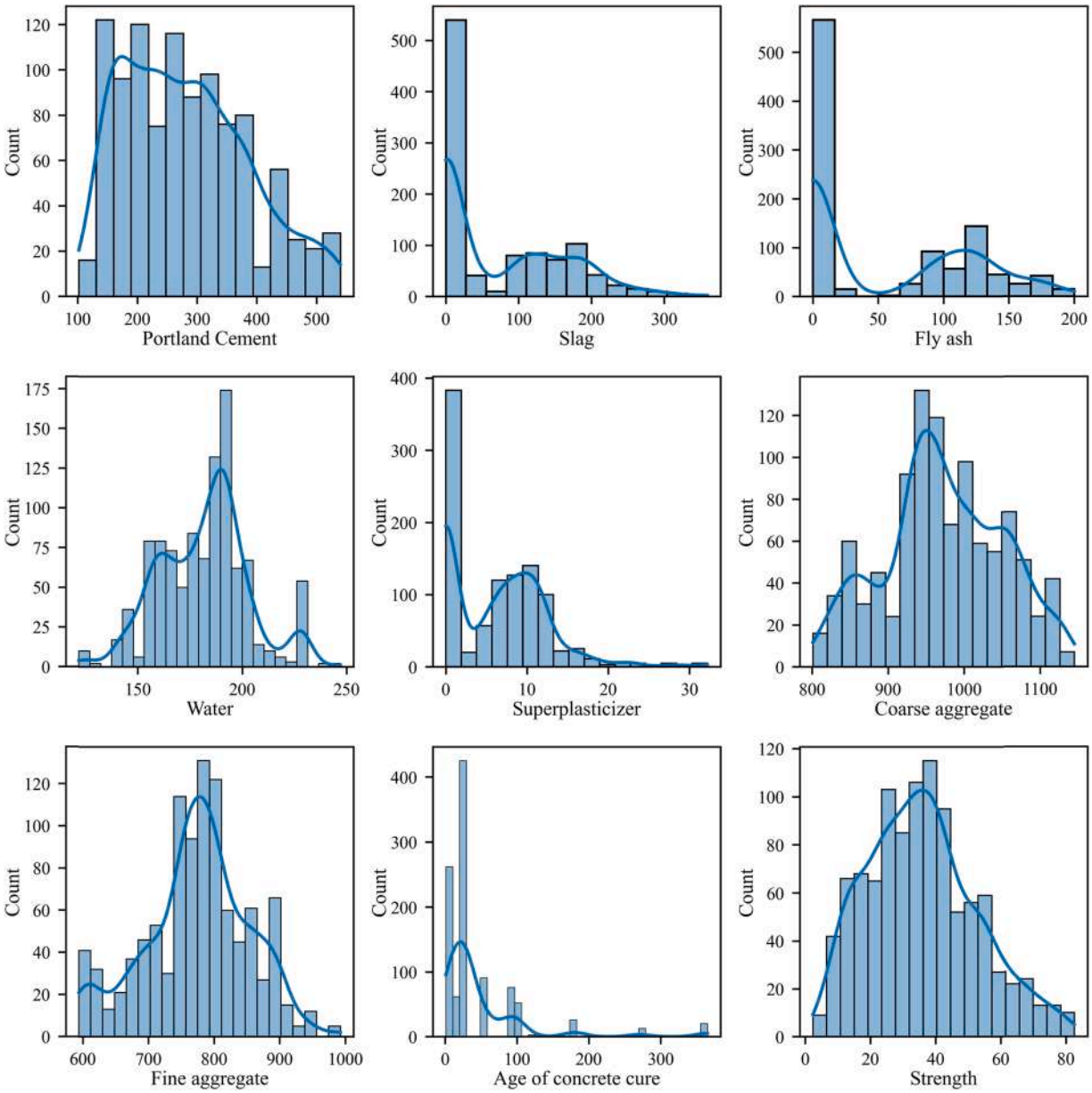

Fig. 2. Distributions of input parameters and concrete compressive strength.

$$R^2 = 1 - \frac{\sum_{i=1}^{j}(P_i - T_i)^2}{\sum_{i=1}^{j}(T_i - \overline{T})^2} \tag{11}$$

Root mean squared error (RMSE)

$$RMSE = \sqrt{\frac{\sum_{i=1}^{j}(P_i - T_i)^2}{j}} \tag{12}$$

Mean Absolute Error (MAE)

$$MAE = \frac{\sum_{i=1}^{j}|P_i - T_i|}{j} \tag{13}$$

where, T_i and P_i is the experimental compressive strength and the predicted compressive strength, respectively; $\overline{T}$ is the mean experimental compressive strength; j is the number of samples.

The dataset is partitioned into two segments, with 80% of the data used for training the model and 20% for validating the prediction accuracy of the model. During the model training process, a 10-fold cross-validation method is employed to further ensure the performance of the model. During the training process, Gaussian optimization is applied to solve the CASH problem. Subsequently, the predictive performance of the ML models employed was analyzed as a comparison. Finally, for the EBM model, its global and local interpretability is discussed. The flow chart of the research procedure is shown in Fig. 3.

6. Implementation of the predictive model

6.1. Bayesian optimization procedure

The implementation of ML-based predictive models for concrete compressive strength prediction is treated as a CASH problem. The

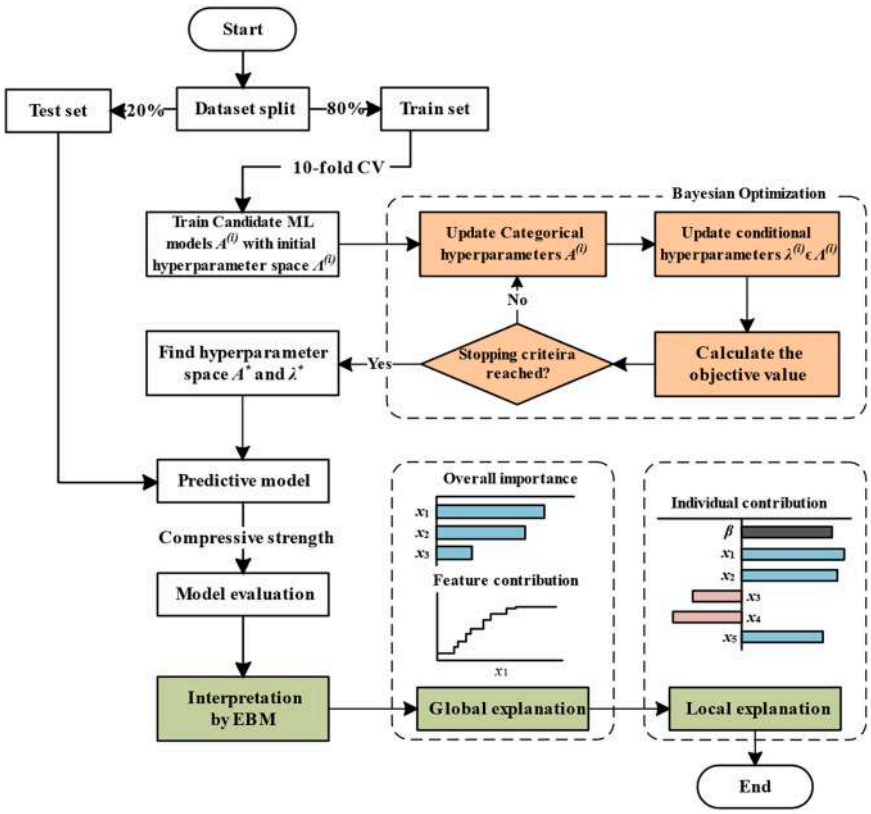

Fig. 3. Schematic diagram of the research process.

problem is treated as a single combined hierarchical hyperparameter optimization problem with model selection included in the hyperparameter space. It is solved by a fully automated approach, leveraging Bayesian optimization for conditional hyperparameter optimization [50]. During the Bayesian optimization, 10-fold cross-validation is utilized, where 9 folds are for training and left 1-fold for validation. The objective metric is set to be R^2 for all models, with the number of iterations set to 100. The hyperparameter search space consists of the model selection, including DT, RF, XGBoost, and EBM, and hyperparameters for the conditional model.

The optimization history of R^2 is shown in Fig. 4. The initial R^2 value is low, and as the optimization progresses, the value of TPE of Bayesian optimization for the hyperparameter makes the model performance tend to be better. The best R^2 result is 0.9262 in the validation set. The hyperparameter importance of the EBM is shown in Fig. 5. The hyperparameter importance are calculated by Hutter [24]. All the importance values are summed up to 1.0. The most important hyperparameter of EBM is max leaves. Interactions of input features rank second. The range of the important hyperparameters is set iteratively to ensure including the best values. With the aid of hyperparameter importance, users could pay more attention to set the ranges of the relatively more import hyperparameters. It will greatly help reduce time consumption and manual intervention.

The hyperparameter optimization process for 100 iterations is shown in Fig. 6. Due to space limitations, only the model and five most important features of EBM are listed in the figure. The principles of the optimization process of other models and hyperparameters are similar. It should be noted that, compared with grid search, in the optimization process of Bayesian optimization, the value of a single hyperparameter does not increase in one direction, nor can it be drawn into a curve like the literature [48,51]. The value of a single hyperparameter is determined by TPE, and the value that increases EI the most is chosen.

The iteration steps are denoted by color. From Fig. 6(a), the candidate models are viewed as a conditional hyperparameter to others. As the iteration progresses, the model selection is more biased towards EBM, because EBM is more likely to obtain higher R^2 values than XGBoost, RF and DT. The other subgraphs show the optimization process of other hyperparameters. It can be seen that the hyperparameter values at the beginning of the iteration are relatively discrete. In the later stages of the iteration, the hyperparameter values are gradually concentrated in the area that maximizes the objective value improvement. Compared with grid search and random search, this kind of Bayesian optimization based on prior information can effectively reduce the search time of hyperparameters, reduce manual intervention, and can obtain more accurate results.

6.2. Comparison with other hyperparameter optimization methods

To further validate the efficiency and accuracy of Bayesian optimization, the hyperparameter optimization results are compared with grid search and random search. Table 2 illustrates the search range of hyperparameters of EBM. Due to limited space, only the HPO of EBM are presented here. With grid search, all hyperparameter set up to 6250 are tested and the one yielding the best R^2 is picked out as the final set. The iterations of random search and Bayesian optimization are set to 100.

The final metric of each method and the time consumed are presented in Table 3. The search reaches the highest R^2 of 0.9284. Despite this, it needs 2500 trials and obtain the optimal hyperparameters at the 1481st iteration, which is much slower than the other two methods. The random search requires only 78 trails to obtain optimal hyperparameters within 100 trails. Although being the fastest, it gets the lowest R^2. It took only 63 iterations for the Bayesian optimization to reach the highest R^2, which is much less than grid search. Since it is an informed search method, it needs fewer trials than random search within the same number of trials. Overall, Bayesian optimization is the best approach to consistently attain the optimal hyperparameters among these three methods.

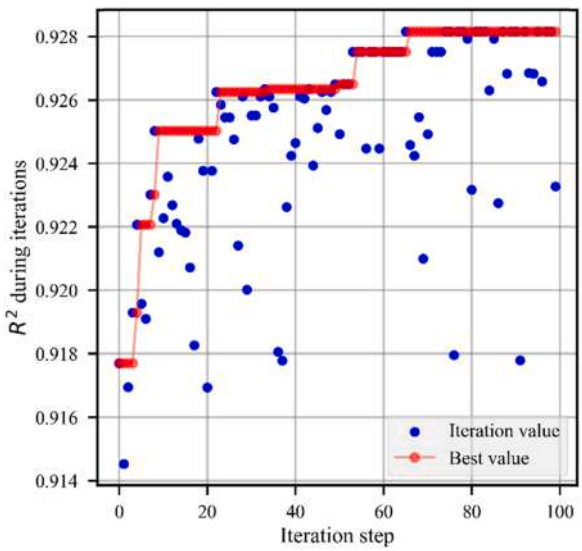

Fig. 4. Hyperparameter importance of EBM.

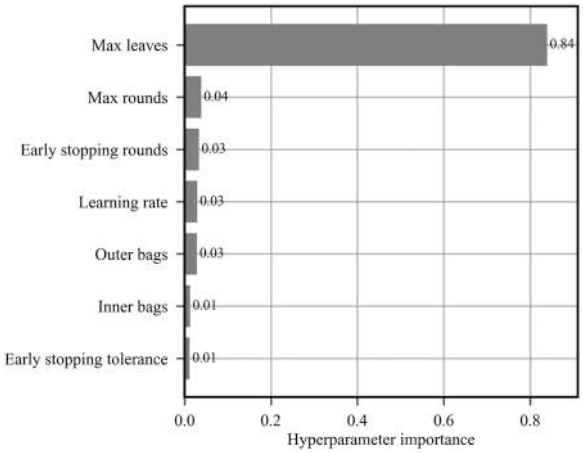

Fig. 5. R^2 history of Bayesian optimization.

7. Results and discussion

7.1. Performance evaluation

Fig. 7 shows the results of EBM and other machine learning models in predicting concrete compressive strength, with the experimental and predicted values on the X and Y axes, respectively. Compared to the DTs model, the integrated tree models have significantly higher prediction accuracy, which is consistent with previous findings [17]. As for the three integrated tree models, RF, XGBoost, and EBM, their scatter points almost overlap with the ideal $y = x$ line, indicating that the predicted and tested values are almost the same. The EBM exhibits predicted values that are more consistent with the experimental values between 10 and 40 MPa in compressive strength compared to the other ML models.

The EBM algorithm provides an excellent performance in predicting compressive strength, and its R^2 is 0.93, which is slightly higher than the 0.91 of the XGB and 0.88 of the RF algorithms, as shown in Fig. 8. This indicates that the prediction results of the EBM algorithm have a high correlation. The DT algorithm, as a weak learner, has only 0.66 in R^2. It is obvious that the method based only on simple classification cannot make an effective prediction of concrete compressive strength under multifactor influence. According to the results of *RMSE* and *MAE*, the excellent prediction performance of EBM is also demonstrated with a small error value with *RMSE*= 4.33 and *MAE*= 3.10. Table 4 provides detailed illustrations of the statistical characteristics of the model errors and prediction metrics.

7.2. Model error distribution

The box plot of the error distribution for the four ML models is shown in Fig. 9. The plot shows the interquartile range (IQR) with the mean and median values of the predicted compressive strength. Outliers are defined as data points that exceed over 1.5 times the IQR. In terms of the dispersion of the prediction results, EBM exhibits the smallest IQR, while DT has the largest dispersion. With respect to model prediction accuracy, the average values of EBM and XGBoost are very close to the perfect prediction of 1.0. It indicated that EBM provides a high level of precision and reliability for predicting the compressive strength of concrete.

7.3. Importance of global characteristics

The EBM algorithm is a generalized additivity model based on the tree-based model. The contribution of features may be graded and plotted to show the impact on individual prediction in both global and local aspects due to additivity. The global explanation of the EBM allows to visualize the effect of each combination of parameters on the predicted results of compressive strength. Fig. 10 shows a summary of the global importance of each combination parameter, which demonstrates the importance of all combination factors. It can be seen that the age of curing and the content of cement play a decisive role in the compressive strength. The concrete hardening process is a continuous process of hydration reaction between cement and other cementitious materials, such as water. Among them, cement, as the main cementitious material, plays a decisive role in the growth of the strength. The influence of the age of curing on the compressive strength of concrete is equally important. Adequate curing time enables the cement to be hydrated more adequately, resulting in higher compressive strength. In some calculation models that describe the time-dependent growth of the compressive strength of concrete, such as the FIB model and the ACI model, it can be found that the compressive strength of concrete will show a tendency to grow faster and then slower with time. The next major influences on compressive strength are slag, water and fly ash, which contain the necessary elements involved in the hydration reaction and common supplementary cementitious materials. Aggregates and superplasticizers have a lower impact on the strength because they are largely not involved in chemical reactions.

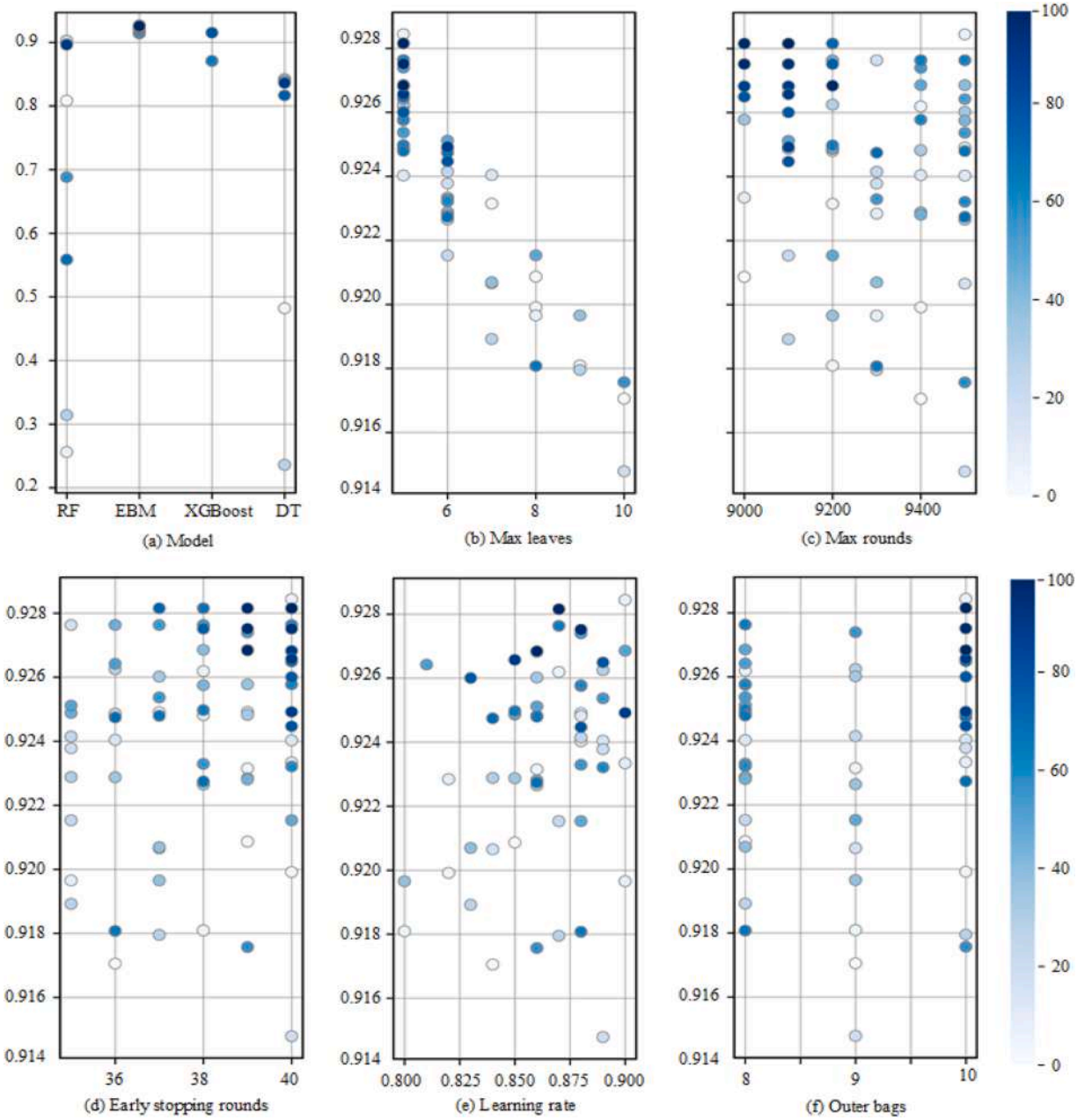

Fig. 6. Bayesian hyperparameter optimization process for 100 iterations.

Table 2
Hyperparameter configurations for EBM.

Hyperparameter	[start, stop, step]	Number of points	Best value
Max leaves	[5, 10, 1]	5	5
Max rounds	[9000, 9500, 100]	5	9500
Early stopping rounds	[35, 40, 1]	5	39
Learning rate	[0.80, 0.90, 0.01]	10	0.87
Outer bags	[8, 10, 1]	2	10

Table 3
Hyperparameter optimization results of different methods.

Optimization strategy	Number of iterations	Index of optimal hyperparameters	Best R^2	Time elapsed (s)
Grid search	2500	1481	0.9284	8367
Random search	100	78	0.8992	274
Bayesian optimization	100	63	0.9282	376

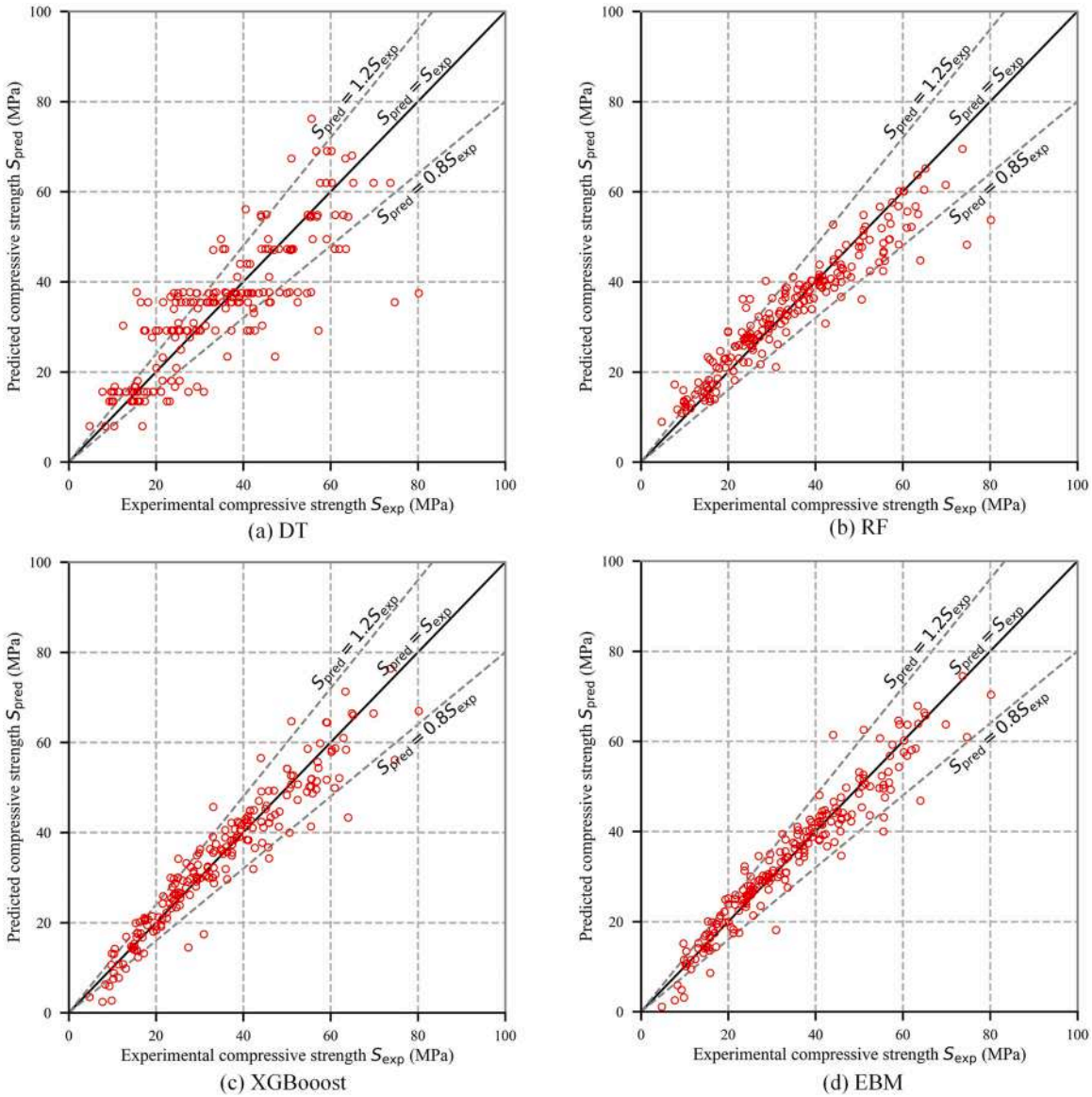

Fig. 7. Scatter plot of predicted versus actual values.

7.4. Evaluation accuracy in various features

The feature shape functions of the EBM allow showing the quantified contribution of individual features to the individual specimen. EBM maintains high intelligibility because the contribution of each feature to an individual's final prediction can be quantified. In Eq. (7), each shape function is a complex non-linear function obtained by bagging and gradient boosting. By restricting the boosting procedure to train on one feature at a time in a round-robin fashion with a very low learning rate, the order of features does not matter. To mitigate the effect of co-linearity, the model rounds-robin cycles through features to learn the best feature shape function for each one. It shows how each feature contributes to the model's prediction. Finally, the individual final prediction is obtained by summing up

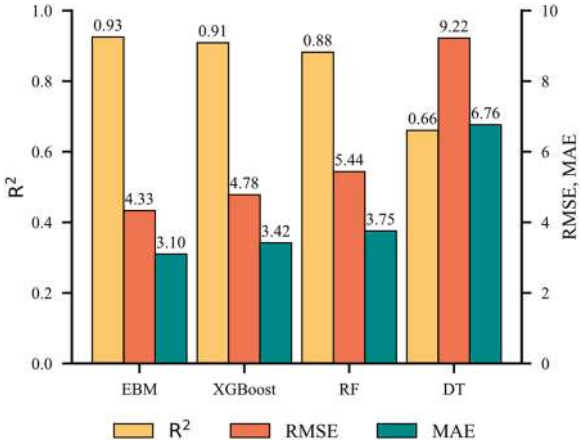

Fig. 8. Prediction performance metrics of the ML models.

Table 4
Statistical properties of ML prediction results.

Model	R^2	*RMSE*	*MAE*	μ_M	σ_M	*COV*
EBM	0.925	4.33	3.10	1.00	0.16	0.16
XGBoost	0.909	4.78	3.42	1.00	0.16	0.16
RF	0.882	5.44	3.75	1.04	0.19	0.19
DT	0.661	9.22	6.76	1.06	0.32	0.30

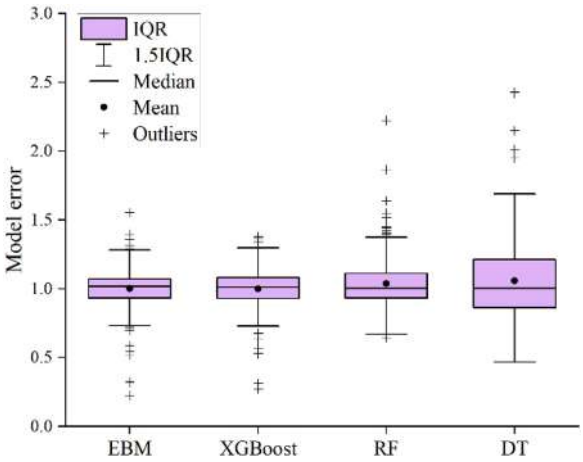

Fig. 9. Prediction error distribution of the ML models.

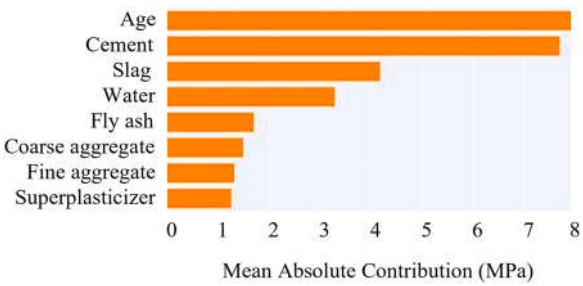

Fig. 10. Overall ranking of importance of features.

all the feature shape functions at given features. The contribution values of all variables to the prediction results are given in Fig. 11. For each subfigure, the top line plot shows the contribution of features, where the gray band is the predicted accuracy. The bar chart below corresponds to the data density of that variable. By analyzing the trends of the line graphs of individual variables, the influence of each ratio factor on the compressive strength of concrete can be analyzed. Selected variables that have a greater impact weight on the predicted results of compressive strength are discussed below.

The effect of the amount of cement mixture on the compressive strength is shown in Fig. 11(a), where it is seen that the compressive strength of the concrete tends to increase linearly as the amount of cement mixture increases. Cement is the main cementitious material that provides strength in concrete and is hardened by the hydration reaction of cement. The more cement that is mixed, the more cementite that will be produced after the reaction, while concrete has higher strength. The admixture of slag provides a certain increase in strength, although the increase is limited. Furthermore, the growth of compressive strength does not increase with increasing slag admixture, as shown in Fig. 11(c). Indeed, the slag has a certain chemical activity and can participate in the hydration reaction. Moreover, slag can have a filling effect on concrete, thus improving the microstructure of concrete paste. Therefore, slag is generally considered a good alternative material to cement, with high economic and environmental properties. Water is one of the elements necessary for the cement hydration reaction, and through Fig. 10(d), the effect of water on compressive strength first increases and then decreases. It is well known that the water-cement ratio has a great effect on the compressive strength of concrete, and the proper amount of water admixture can ensure that the desired concrete strength is obtained. If the water admixture is insufficient, it will lead to inadequate hydration of cement, and if the amount of water is excessive, it will lead to concrete segregation and other problems. From the results in Fig. 11(d), it can be seen that the water admixture has a positive effect on the strength and reaches the highest contribution at 143–153 kg/m^3. Subsequently, the compressive strength of concrete decreases with the increase of the amount of water mixing. Fig. 11(h) shows the effect of the curing age on the predicted results in compressive strength. A wide distribution of curing age is adopted to improve the generalizability of the model. The compressive strength of concrete increases rapidly in the interval of 0–40, slightly larger than 28 due to the feature distribution in the train set. It then slows down with increasing age of cure. This is consistent with our understanding of the development of the compressive strength of concrete. The positive and negative values of the contribution of compressive strength in the vertical coordinate are determined according to the properties of the training data set sample. The positive and negative values represent the contribution to the prediction results, with the positive part providing a positive effect on the strength prediction and the negative part the opposite. When the days of curing was around 22 days, the curing age showed a crossover of positive and negative predicted values for compressive strength. In summary, the EBM algorithm can clearly explain the contribution values of individual characteristics to the predicted results, by which the effect of each mix ratio parameter on the compressive strength of concrete can be further determined.

7.5. Local explanation

The EBM algorithm allows to give detailed contributions of variables for a single prediction. As an example, Fig. 12 shows the results of a typical individual compressive strength prediction. The predicted compressive strength is 44.0 MPa, which is slightly lower than the experimental value of 44.6 MPa. In terms of the contribution of each variable to the predicted compressive strength results, cement had a negative effect on the predicted results, while all other variables had a positive effect. The examination of the mix proportion makes it clear that the cement content is low (142.0 Kg/m^3) and is supplemented by supplementary cementitious materials of slag and fly ash. As this reduces the mixing amount of active components in the concrete and increases the W/C, which resulting in reducing the contribution of cement to the predicted result. EBM-based interpretability plays an important role in concrete compressive strength prediction. It exhibits excellent prediction performance while assigning interpretability to the output results. Being able to provide assistance in the design of concrete mix, it has good potential for application.

8. Conclusions

To investigate the effect of mixing ratios of different raw materials on the compressive strength of concrete, a machine learning (ML) predictive model based on Explainable Boosting Machine (EBM) is proposed to predict the compressive strength of concrete. The Bayesian optimization algorithm is also applied to solve the problem of model selection and hyperparameter optimization during ML model building. The prediction performance of four ML models, decision trees (DTs), random forest (RF), XGBoost and EBM, was verified with 1030 sets of experimental data of the concrete compressive strength. The global and local output results of the EBM model revealed the patterns of each mix ratio factor that affect the compressive strength of the concrete, which in turn verified the interpretability of the model. The main conclusions are as follows.

1. A predictive model for concrete compressive strength based on EBM is proposed in this study. It maintains inherit interpretability while acquiring high prediction accuracy. The complex shape function obtained by gradient boosting guarantees the accuracy of the model, and the interpretability lies in the fact that the final prediction is the sum of individual shape functions. Interpretability reveals the decision-making process of EBM and quantifies how each feature contributes to the final prediction.
2. Bayesian optimization identify optimal points in the combined algorithmic/hyperparametric space automatically, significantly reducing the time consumption and manual intervention. Compared to grid search and random search, Bayesian optimization consistently achieves optimal hyperparameters with fewer iterations.

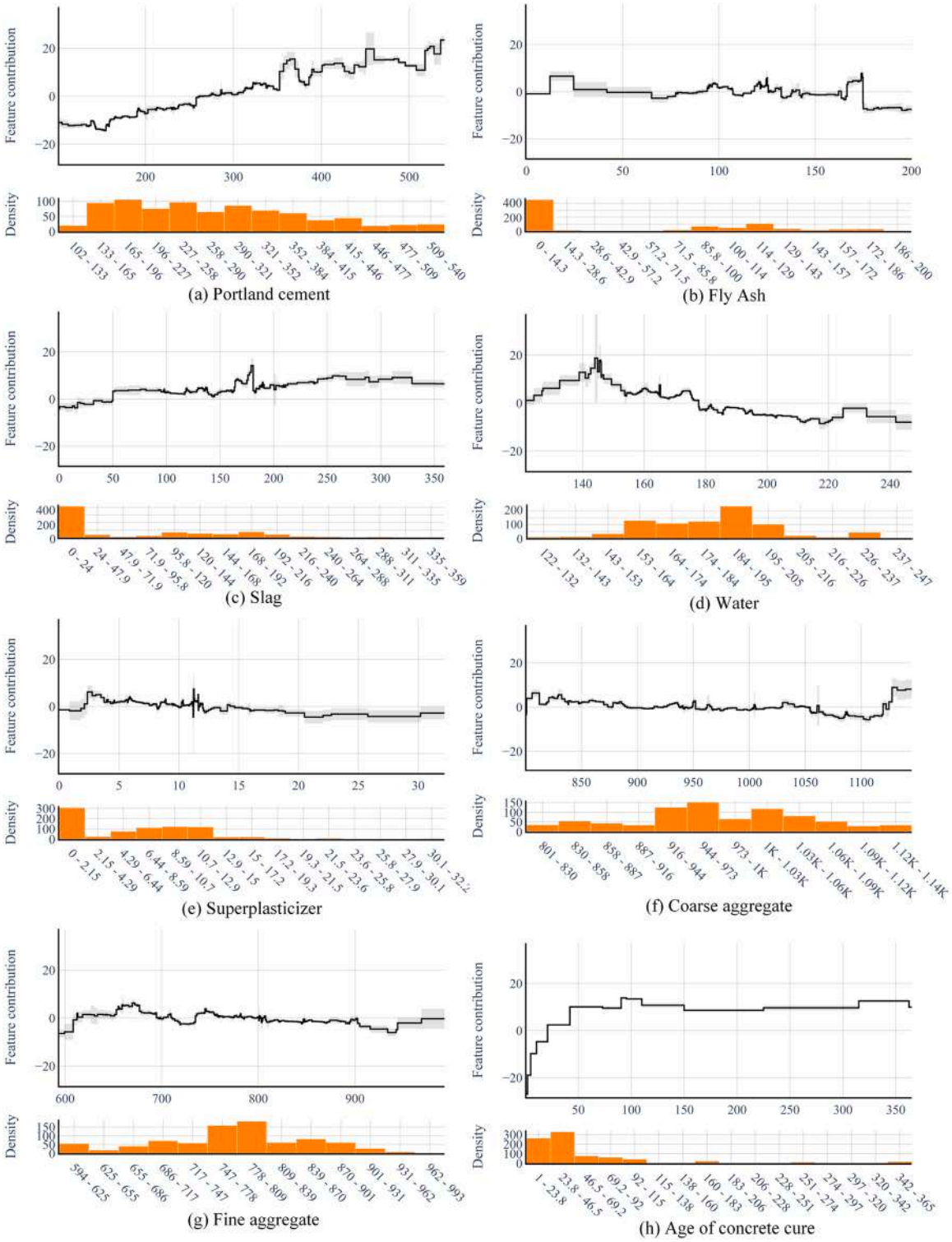

Fig. 11. Global explanation of the contribution value of individual characteristics.

3. Among the four applied ML models, the EBM model had the best predictive performance with R^2= 0.93, *RMSE*= 4.33, *MAE*= 3.10. In terms of the dispersion of the prediction results, EBM exhibits the smallest IQR, while DT has the largest dispersion. The average values of EBM and XGBoost are very close to the perfect prediction of 1.0.

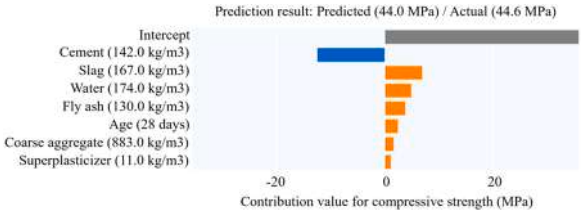

Fig. 12. Local explanation of a specific prediction result.

4. Regarding the feature importance, the age of curing and cement admixture exhibit a more significant effect in the prediction of compressive strength. The EBM model is able to provide both global and local interpretations of the compressive strength prediction results. The global interpretation helps to understand the influence pattern of each mixing parameter on the compressive strength. The local interpretation allows to know the actual contribution of each parameter in a certain prediction outcome.

Although the proposed machine learning model exhibits excellent predictive performance for concrete compressive strength prediction, it still has some limitations for utilization. The proposed model is able to provide reliable predictions for each range of input parameters. When the parameters exceed the specific range, the credibility of the period prediction results needs further validation. For data-driven ML methods, the performance is generally affected by the volume and reliability of data. In the future, dataset used in this paper can be increased in the future to improve model's performance. Moreover, the emerging ML methods should be further incorporated and compared with the current method to find a more appropriate model for concrete compressive strength prediction.

CRediT authorship contribution statement

Gaoyang Liu: Investigation, Methodology, Visualization, Writing – original draft. **Bochao Sun:** Experiment, Investigation, Writing – original draft, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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